# §14.6 Leibniz Formula

**Def:** $\vec{\sigma} = (\sigma_1, \sigma_2, \cdots, \sigma_n)$ is a permutation of $(1, 2, \cdots, n)$.

Ex. if $n=3$, $\vec{\sigma}$ can be $(1,2,3), (1,3,2), (2,1,3)$
$(2,3,1), (3,1,2), (3,2,1)$

**Def.** $\tau(\vec{\sigma}) = (-1)^m$, where $m$ is # of swaps it requires in order to change $(1, \cdots, n)$ to $\vec{\sigma}$.

Ex. For $\vec{\sigma} = (1,2,3)$, $\tau(\vec{\sigma}) = (-1)^0 = 1$

Ex. For $\vec{\sigma} = (3,1,2)$

$(1,2,3) \to (2,1,3) \to (3,1,2)$, so $m=2$, $\tau(\vec{\sigma}) = 1$

Ex. For $\vec{\sigma} = (2,1,3)$,

$(1,2,3) \to (2,1,3)$, $m=1$, $\tau(\vec{\sigma}) = -1$

**Def:** ~~$f(\vec{\sigma}) =$~~ For $\vec{\sigma} = (\sigma_1, \cdots, \sigma_n)$ and $A \in \mathbb{R}^{n \times n}$

$$f(\vec{\sigma}) = \tau(\vec{\sigma})\, a_{\sigma_1 1}\, a_{\sigma_2 2} \cdots a_{\sigma_n n}$$

Ex. $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$, $\vec{\sigma} = (2,1,3)$

$$f(\vec{\sigma}) = \tau(\vec{\sigma})\, a_{21} a_{12} a_{33} = -72$$

**Theorem (Leibniz Formula):** $\det(A) = \sum_{\text{all } \vec{\sigma}} f(\vec{\sigma})$

Ex. $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$, $\vec{\sigma} \in \{(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1)\}$

$\vec{\sigma} = (1,2,3)$: $f(\vec{\sigma}) = 1 \times a_{11} \times a_{22} \times a_{33} = 45$

$(1,3,2)$: $f(\vec{\sigma}) = (-1) \times a_{11} \times a_{23} \times a_{32} = -48$

$(2,1,3)$: $f(\vec{\sigma}) = -72$

$(2,3,1)$: $f(\vec{\sigma}) = 1 \times a_{\cancel{21}} \times a_{\cancel{32}} \times a_{\cancel{13}} = \cancel{84}$

$\vec{\sigma} = (3,1,2)$: $f(\vec{\sigma}) = 1 \times a_{31} \times a_{12} \times a_{23} = 96$

$(3,2,1)$: $f(\vec{\sigma}) = (-1) \times a_{31} \times a_{22} \times a_{13} = -105$

$\Rightarrow \det(A) = 45 - 48 - 72 + 84 + 96 - 105 = 0$

Proof: Apply induction.

When $n=1$, $\det(A) = a_{11}$. Clearly true.

Assume that the $n=k$ case is also true

so $\forall A \in \mathbb{R}^{k\times k}$, $\det(A) = \sum_{\vec{\sigma}} f(\vec{\sigma})$

When $n=k+1$, $A = \begin{bmatrix} a_{11} & \cdots & a_{1,k+1} \\ \vdots & & \vdots \\ a_{k+1,1} & \cdots & a_{k+1,k+1} \end{bmatrix}$

Let $Q_1 = \left[\begin{array}{c|c} a_{11} & a_{12} \cdots a_{1,k+1} \\ \hline 0 \\ \vdots \\ 0 \end{array} \right.$ $\left. \begin{array}{c} \\ \hline \\ M_{11} \end{array}\right]$ $Q_2 = \left[\begin{array}{c|c} 0 & a_{12} \cdots a_{1,k+1} \\ \hline a_{21} & a \\ 0 \\ \vdots \\ 0 \end{array}\right]$

Let $Q_1 = \left[\begin{array}{c|c} a_{11} & a_{12} \cdots a_{1n} \\ \hline 0 \\ \vdots \\ 0 \end{array} \right.$ $\left. \begin{array}{c} \\ \hline \\ a_{n2} \cdots a_{nn} \end{array}\right]$, $\cdots$, $Q_n = \left[\begin{array}{c|c} 0 & a_{12} \cdots a_{1n} \\ \hline \vdots \\ 0 \\ a_{n1} \end{array} \right.$ $\left. \begin{array}{c} \\ \hline \\ a_{n2} \cdots a_{nn} \end{array}\right]$

Then $\det(A) = \det(Q_1) + \cdots + \det(Q_n)$

where $\det(Q_i) = a_{i1} A_{i1} = (-1)^{i+1} a_{i1} \det(M_{i1})$

since $M_{i1} \in \mathbb{R}^{k\times k}$, $= (-1)^{i+1} a_{i1} \cdot \sum_{\vec{\sigma}'} f(\vec{\sigma}')$,

$\vec{\sigma}' = (\sigma_2, \cdots, \sigma_n)$

Let $\vec{\sigma} = (\sigma_1, \sigma_2, \cdots, \sigma_n)$

$f(\vec{\sigma}) = \tau(\vec{\sigma}) a_{\sigma_1 1} a_{\sigma_2 2} \cdots a_{\sigma_n n}$

if $\sigma_1 = i$, then $\tau(\vec{\sigma}') = \tau(\sigma_2, \cdots, \sigma_n) = \tau(\sigma_2, \cdots, \underset{\text{ith col}}{i}, \cdots, \sigma_n)$

$= (-1)^{i-1} \tau(i, \sigma_2, \cdots, \sigma_n) = (-1)^{i-1} \tau(\vec{\sigma})$

$\Rightarrow f(\vec{\sigma}) = (-1)^{1-i} \tau(\vec{\sigma}') a_{\sigma_1 1} \cdots a_{\sigma_n n} = (-1)^{i+1} a_{i1} f(\vec{\sigma}')$

$\Rightarrow \det(Q_i) = (-1)^{i+1} a_{i1} \cdot \sum_{\vec{\sigma}'} f(\vec{\sigma}') = \sum_{\vec{\sigma}} f(\vec{\sigma})$